

INTERNSHIP REPORT

By

JANGID ANIL RAMJILAL

202201103142



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

JAWAHARLAL NEHRU ENGINEERING COLLEGE

2025-26

INTERNSHIP REPORT

submitted in partial fulfillment of the requirements for

the award of the degree of

Bachelor of Technology

in

Computer Science and Engineering

By

JANGID ANIL RAMJILAL

202201103142

Under guidance of

Dr. Sandip S. Kankal



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

JAWAHARLAL NEHRU ENGINEERING COLLEGE,

2025-26

Jawaharlal Nehru Engineering College Chhatrapati Sambhajinagar
Department of Computer Science and Engineering



This is to certify that the Internship Report submitted by **JANGID ANIL RAMJILAL** having **PRN 202201103142** is his work during the internship and submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** in **COMPUTER SCIENCE AND ENGINEERING** of MGM University, Chhatrapati Sambhajinagar during the academic year 2025-26.

GUIDE

DR. SANDIP S. KANKAL

HOD

**DR. D. S.
DESHPANDE**

PRINCIPAL

**Dr. V. B.
MUSANDE**



InterviewBit Software Services Private Limited
(Earlier referred to as InterviewBit Software Services LLP)
CIN: U74999KA2021PTC176674

Registered Address: Surya Park – II, 5th Floor, Survey Nos. Sy No 91/1, 91/2, 91/3,
Veerasandra Village, Hosur Road, Bangalore South,
Electronics City, Bangalore – 560 100, Karnataka, India.
Karnataka Vartur Bangalore South Bangalore KA 560087

Date- 18th April 2026

To Whomsoever It May Concern

This is to inform that Anil Jangid has been working as a Quality Assurance Intern (Employee code-**8622**) with InterviewBit Software Service Private Limited (earlier referred to as “InterviewBit Software Services LLP) from **16th December 2025**.

InterviewBit Software Services Private Limited

(earlier referred to as “InterviewBit Software Services LLP)

Surya Park-II, 5th floor, Survey Nos. Sy No 91/1, 92/2, 91/3, Veerasandra Village, Hosur Road,
Electronic City, Bangalore- 560100

His Current Address as per our records:

Raghuram PG for Gents, Gollahalli, Balaji Layout Road, Electronic City,
Bengaluru -560100

Feel free to reach out to us in case of any queries.

Sincerely,

For InterviewBit Software Service Private Limited

Sneha Sindhe
Director (Human Resources)
Ph: +91-9964972031

Corporate Address:Surya Park – II, 5th Floor, Survey Nos. Sy No 91/1, 91/2, 91/3, Veerasandra Village, Hosur Road,
Bangalore South, Electronics City, Bangalore – 560 100, Karnataka, India, Karnataka Vartur Bangalore South Bangalore
KA 560087

ACKNOWLEDGEMENT

I extend my heartfelt appreciation to all those who played a pivotal role in the successful culmination of my internship report.

Firstly, I express my sincere gratitude to Dr. Sandip S. Kankal, my esteemed mentor at Jawaharlal Nehru Engineering College, for his unwavering support and invaluable guidance throughout this internship endeavor. His profound insights and encouragement have significantly shaped my understanding and approach to the subject matter.

I am deeply thankful to the entire team at Scaler AI Labs (InterviewBit Software Services Private Limited) for granting me the opportunity to embark on this enriching journey as a QA Intern. Special acknowledgment goes to my supervisors and the development team for their consistent guidance, patience, and willingness to impart their expertise. Their mentorship has not only honed my technical skills in quality assurance but has also broadened my perspective on the practical applications of manual testing in AI-driven real-world environments.

Additionally, I express my gratitude to the faculty members of Jawaharlal Nehru Engineering College for their continuous support and encouragement throughout my academic tenure. Their invaluable teachings have laid a robust foundation for my professional development and growth.

The completion of this report is a testament to the collective efforts and support of these individuals and organizations, and I am genuinely appreciative of their contributions.

CONTENTS

	List of Abbreviations	i
	Abstract	ii
1.	COMPANY PROFILE	1
1.1	Organization Structure	1
1.2	Employees	2
1.3	Products and Services	2
1.4	Achievements	3
1.5	Departments	4
2.	INTERNSHIP DISCUSSION	5
2.1	Work Assignments	5
2.2	Objectives	8
2.3	Plan	11
2.4	Progress	12
2.5	Acquiring and Application of Knowledge	14
2.6	Use of Modern Tools and Software	16
2.7	Use of Management Practices	17
2.8	Design/Development of Solutions	19
2.9	Completion Status	21
3.	WEEKLY OVERVIEW	22
4.	CONCLUSIONS	24
	References	iii

List of Abbreviations

Sr No.	Abbreviation	Full Form
1.	QA	Quality Assurance
2.	RL	Reinforcement Learning
3.	AI	Artificial Intelligence
4.	ML	Machine Learning
5.	UI	User Interface
6.	API	Application Programming Interface
7.	STLC	Software Testing Life Cycle
8.	BRD	Bug Report Document

ABSTRACT

This internship report encapsulates the enriching experience garnered during the tenure as a Quality Assurance (QA) Intern at Scaler AI Labs (InterviewBit Software Services Private Limited) — an innovative organization that teaches AI to do real work. The report provides a comprehensive overview of the internship objectives, activities, and the knowledge and skills acquired throughout the internship period, which spanned from 16th December 2025 to 31st May 2026.

The primary focus of the internship was the manual testing of Reinforcement Learning (RL) environments developed by the organization — high-fidelity replica web applications simulating real-world platforms. Testing was carried out at a pace of one to two RL environments per week, and a total of 12 environments have been successfully QA-tested across diverse domains including document management (Adobe Sign), scheduling (Cal.com, Google Calendar), pet services (Rover), collaboration (MS Teams), recruitment (Workable), project management (Trello), e-commerce (Shopify), social media (Buffer), cloud storage (Box), accounting (QuickBooks), and enterprise resource planning (SAP). This involved thorough understanding of each environment's expected behavior, systematic identification of defects through structured test execution, and meticulous documentation of bugs in an Excel-based bug tracking sheet. Post bug-fix, the intern was also responsible for verifying the resolution of reported issues and providing final QA sign-off for each environment.

Challenges encountered during the internship, such as understanding the complex functional flow of RL environments and effectively communicating technical intricacies to the development team, are candidly

addressed in this report. Strategies employed to surmount these challenges, including collaborative problem-solving, domain research, and active mentorship, are elaborated to provide a holistic picture of the internship journey.

Reflections on personal and professional growth form an integral part of this report. The internship catalyzed the enhancement of technical proficiency in QA methodologies, as well as the development of soft skills such as analytical thinking, communication, and adaptability. In conclusion, the internship at Scaler AI Labs (InterviewBit Software Services Private Limited) proved to be a transformative journey, characterized by invaluable learning experiences and meaningful contributions to the quality assurance of cutting-edge AI systems.

1. COMPANY PROFILE

Scaler AI Labs (InterviewBit Software Services Private Limited) is an innovative technology organization operating under the philosophy "we teach AI to do real work." Incorporated under the name InterviewBit Software Services Private Limited (CIN: U74999KA2021PTC176674), the company operates its AI research and development division under the brand Scaler AI Labs. With its registered address at Surya Park-II, 5th Floor, Electronics City, Bangalore – 560100, the organization focuses on developing intelligent systems capable of performing complex, meaningful tasks autonomously. It specializes in building and evaluating Reinforcement Learning (RL) environments — high-fidelity replica environments of real-world applications — that serve as training and testing grounds for AI agents.

With a strong commitment to AI research, quality, and innovation, Scaler AI Labs (InterviewBit Software Services Private Limited) has positioned itself at the forefront of applied artificial intelligence. The organization fosters a dynamic environment where engineers, researchers, and quality assurance professionals work collaboratively to ensure that the AI systems developed are robust, reliable, and aligned with real-world requirements.

1.1 Organization Structure

Scaler AI Labs (InterviewBit Software Services Private Limited) follows a flat, agile organizational structure that promotes cross-functional collaboration and rapid iteration. The organization comprises top-level leadership, research scientists, software engineers, AI/ML engineers, and quality assurance professionals. This structure ensures efficient communication and swift decision-making across all levels.

The QA team, to which the intern was assigned, operates in close coordination with the development and research teams. This close-knit collaboration ensures that quality standards are upheld throughout the development lifecycle of RL environments.

1.2 Employees

Scaler AI Labs (InterviewBit Software Services Private Limited) values a diverse and skilled workforce comprising talented professionals across domains such as machine learning, software engineering, quality assurance, product management, and data science. The organization places a strong emphasis on continuous learning and professional development, providing employees and interns with opportunities to work on challenging, real-world AI problems.

Interns at Scaler AI Labs (InterviewBit Software Services Private Limited) are treated as integral members of the team, given meaningful responsibilities, and are encouraged to contribute actively to ongoing projects. The collaborative culture ensures that knowledge transfer and mentorship are seamlessly embedded into the day-to-day work experience.

1.3 Products and Services

Scaler AI Labs (InterviewBit Software Services Private Limited) develops a suite of AI-driven products and services centered around Reinforcement Learning and intelligent automation:

1. **RL Environment Development:** Design and development of custom Reinforcement Learning environments that simulate real-world tasks, enabling AI agents to learn optimal decision-making strategies through interaction and feedback.

2. **AI Agent Training Platforms:** Scalable platforms that facilitate the training of AI agents across diverse domains, ranging from task automation to complex problem-solving scenarios.
3. **Quality Assurance of AI Systems:** Systematic evaluation and testing of AI models and environments to ensure correctness, reliability, and alignment with intended behavior and performance benchmarks.
4. **Research and Experimentation:** Conducting cutting-edge research in reinforcement learning, reward modeling, and AI evaluation methodologies to continuously advance the state of the art.
5. **Consulting and Custom AI Solutions:** Providing tailored AI solutions to organizations seeking to integrate intelligent automation and RL-based systems into their operations.

1.4 Achievements

Scaler AI Labs (InterviewBit Software Services Private Limited) has achieved significant milestones in its journey to make AI capable of doing real work. The organization has successfully developed and deployed multiple Reinforcement Learning environments that are used for training and benchmarking AI agents across a variety of real-world task categories. Its commitment to quality-first AI development has been recognized within the AI research community, with ongoing collaborations with academic institutions and industry partners.

The organization continues to expand its research portfolio, with a focus on scalable RL training, environment fidelity, and robust QA frameworks tailored specifically for AI-driven systems.

1.5 Departments

Scaler AI Labs (InterviewBit Software Services Private Limited) operates through several key departments, each playing a vital role in delivering high-quality AI products and services:

Research and Development: Focused on advancing the theoretical and applied foundations of Reinforcement Learning and AI, this department drives innovation across all product lines.

Engineering: Responsible for designing, building, and maintaining the software infrastructure, RL environment codebase, and AI agent training pipelines.

Quality Assurance: Ensures the correctness and reliability of RL environments and AI systems through systematic manual and automated testing, bug reporting, and verification processes.

Product Management: Oversees the roadmap, prioritization, and delivery of AI products, ensuring alignment with organizational goals and user requirements.

2. INTERNSHIP DISCUSSION

2.1 Work Assignments

During the internship at Scaler AI Labs (InterviewBit Software Services Private Limited), work assignments were structured to provide a comprehensive understanding of quality assurance as applied to Reinforcement Learning environments. These environments are high-fidelity replica web applications simulating real-world platforms across diverse domains such as document management, scheduling, recruitment, project management, e-commerce, social media, finance, and enterprise resource planning. The internship commenced with an onboarding phase to familiarize the intern with the organization's environment architecture, testing approach, and QA processes.

Training Phase:

The initial weeks of the internship were dedicated to understanding the fundamental concepts of RL environments, including how replica applications function, the role of task evaluation logic, expected agent behavior, and how to compare replica environments against original real-world applications. This grounding was essential to performing meaningful quality assurance on each assigned environment.

Projects and RL Environments Tested:

Over the course of the internship, a total of 12 RL environments were assigned and successfully QA-tested, progressing at a pace of one to two environments per week. Each environment required independent exploration, functional understanding, and thorough testing before sign-off. The following environments were worked on:

6. Adobe Sign: A document signing and workflow automation platform. Testing involved validating document upload flows, signature request workflows, signing sequences, status tracking, and email notification triggers. Edge cases around multi-party signing and document expiry were also verified.
7. Cal.com: An open-source scheduling and calendar management platform. Testing covered booking workflows, availability configuration, time zone handling, meeting types, and notification behavior. Recurring scheduling logic and edge cases around overlapping slots were thoroughly examined.
8. Rover: A platform simulating pet services and booking workflows. Testing focused on service listing behavior, booking flows, provider-client interactions, messaging workflows, and payment-related scenarios.
9. MS Teams: A collaboration and communication platform replica. Testing included channel creation, messaging workflows, file sharing behavior, meeting scheduling, notification logic, and permission-based access validation across different user roles.
10. Workable: A recruitment and applicant tracking system. Testing covered candidate pipeline workflows, stage transitions, job posting behavior, filters, candidate profile management, messaging workflows, and data consistency across recruitment stages.
11. Trello: A project management and task tracking platform. Testing involved board creation, card management, list transitions, label and checklist behavior, member assignment, due date logic, and activity log validation.
12. Shopify: An e-commerce platform with extensive business logic. Testing covered product and inventory management, order

processing workflows, customer interactions, checkout flows, discount logic, and multiple interconnected edge cases across modules.

13.Buffer: A social media scheduling and content management platform. Testing focused on post scheduling workflows, content queue behavior, platform connection logic, and UI consistency across different publishing states.

14.Box: A cloud content management platform. Testing included file and folder management workflows, sharing and permission logic, version control behavior, collaboration features, and regression testing during final sign-off validation.

15.Google Calendar: A calendar and event management platform. Testing covered event creation, recurring event logic, attendee invitation workflows, time zone behavior, reminders, and date/time-based edge cases including overlapping and multi-day events.

16.QuickBooks: An accounting and financial management platform. Testing included invoice creation and management, expense tracking workflows, payment recording, tax calculation logic, reporting behavior, and data consistency across financial modules.

17.SAP: An enterprise resource planning platform with highly complex business logic. Testing covered multi-module workflows spanning procurement, inventory, order management, and reporting. Extensive validation of data flow between integrated modules and edge case handling was performed.

Bug Documentation and Tracking:

All defects and inconsistencies identified during testing were meticulously documented in a structured Excel-based bug tracking sheet. Each bug report included details such as the bug description, steps to reproduce,

expected behavior, actual behavior, severity classification, and current resolution status. This systematic approach ensured that all reported issues were traceable and actionable by the development team.

Bug Verification:

Upon receiving notifications from developers regarding bug fixes, the intern performed regression testing to verify that the reported issues had been resolved correctly and that no new defects had been introduced as a result of the fixes. The status of each bug in the tracking sheet was updated accordingly, and a final QA sign-off was provided for each environment upon successful completion.

Significance and Relevance:

Each environment tested was directly used in the training and evaluation of AI agents. Ensuring the correctness and reliability of these replica environments was therefore critical — any defect left unresolved could lead to incorrect agent behavior or flawed evaluation outcomes. The intern's contributions directly supported the quality and integrity of the AI systems being developed by the organization.

Challenges and Strategies:

A significant challenge encountered during the internship was the need to rapidly understand the full functional scope of each new environment within a short timeframe. Each platform belonged to a completely different domain, requiring the intern to become a proficient user of the original real-world application before meaningful testing could begin. This was addressed through dedicated self-study of each platform, exploratory testing, and consultation with the development team.

Another challenge was effectively communicating identified bugs to developers in a clear and unambiguous manner, especially when the defect

involved complex multi-step workflows or environment-specific behavior. This was addressed through structured, detailed bug reports supplemented with clear reproduction steps and expected versus actual behavior documentation.

Reflections:

The internship experience has been highly rewarding, providing the opportunity to work across twelve diverse real-world platform replicas and develop a broad and deep understanding of QA across multiple domains. The combination of structured learning, exploratory testing, and collaborative bug resolution provided a well-rounded professional development experience.

2.2 Objectives

- 18.QA Proficiency in AI Environments: To develop and demonstrate proficiency in manual testing of AI and Reinforcement Learning environments, including the ability to design and execute test scenarios that validate environment behavior against specifications.
- 19.Enhancing Problem-Solving Abilities: To strengthen analytical and critical thinking skills by dissecting complex RL environment behaviors, identifying root causes of defects, and proposing effective corrective actions.
- 20.Effective Bug Documentation: To master the art of precise and comprehensive bug reporting, ensuring that all identified defects are documented with sufficient detail for developers to understand, reproduce, and resolve them efficiently.
- 21.Cross-Functional Communication: To develop the ability to communicate complex technical findings clearly and effectively to

both QA peers and software developers, bridging the gap between testing observations and development actions.

22. Understanding RL Concepts: To acquire a foundational understanding of Reinforcement Learning principles, including agent behavior, reward structures, state transitions, and environment design, as a prerequisite for meaningful quality assurance.

23. Ensuring Quality Deliverables: To contribute to the delivery of high-quality, reliable RL environments by systematically identifying defects, verifying fixes, and maintaining comprehensive records of all testing activities.

2.3 Plan

Initial Plan Outline:

The original internship plan was divided into two main phases: an initial learning and onboarding period, followed by hands-on QA engagement with live RL environment projects. The learning phase, scheduled from 16th December 2025 to 9th January 2026, was allocated for domain familiarization, process understanding, and tool onboarding. The practical testing phase was set to commence from 10th January 2026 and continue through 31st May 2026.

Learning Phase (16th December 2025 – 9th January 2026):

- Study RL environment specifications and design documents provided by the team.
- Attend onboarding sessions and walkthroughs conducted by senior team members.
- Familiarize with the Excel-based bug tracking process and reporting standards.

- Conduct exploratory testing on sample environments to build domain understanding.

Practical Testing Phase (10th January 2026 – 31st May 2026):

- Execute systematic manual test cases on assigned RL environments daily.
- Document all identified bugs in the centralized Excel bug tracking sheet.
- Coordinate with developers to clarify environment behavior and verify bug fixes.
- Perform regression testing following developer fixes and update bug statuses.

2.4 Progress

Work Assignment Progress Overview:

Throughout the internship, consistent and measurable progress was made in fulfilling the assigned QA responsibilities. The testing cadence was structured such that one to two RL environments were completed per week, depending on the complexity of the environment and the volume of bugs identified. As of the time of writing this report, more than 15 RL environments have been successfully QA-tested, covering a wide range of task domains, behavioral complexities, and interaction patterns.

Key Achievements:

- Successfully completed QA testing of all 12 assigned RL environments — Adobe Sign, Cal.com, Rover, MS Teams, Workable, Trello, Shopify, Buffer, Box, Google Calendar, QuickBooks, and SAP — at a pace of one to two environments per week, with final sign-off provided for each.

- Identified and documented numerous defects across all tested environments, spanning functional issues, workflow inconsistencies, UI defects, data handling errors, and evaluation logic failures.
- Maintained a well-organized and comprehensive Excel bug tracking sheet that served as the single source of truth for all reported and resolved defects across all 12 environments.
- Effectively communicated complex environment-specific behaviors to developers with clear, structured bug reports, enabling timely and accurate bug resolution.
- Performed thorough regression testing to verify the correctness of all developer-resolved bugs, ensuring no regression in environment behavior after each fix cycle.

Challenges Encountered:

- Each new RL environment presented a unique functional flow and behavioral model, requiring dedicated time at the start of each week to understand the environment's design before testing could begin.
- Balancing the pace of one to two environments per week while maintaining depth and thoroughness of testing required careful time management and prioritization.
- Articulating nuanced RL-specific defects to developers in a clear, actionable manner, especially when the defect involved subtle state-transition or reward-calculation anomalies.
- Managing testing workload efficiently when multiple environments were under concurrent development and required simultaneous QA attention.

Effectiveness of Implemented Strategies:

Time Management: Prioritizing test cases based on the criticality and development stage of each RL environment module ensured that high-risk areas received appropriate testing attention.

Collaboration: Maintaining open and regular communication with the development team facilitated quick clarification of environment specifications and accelerated bug resolution cycles.

Structured Documentation: Adhering to a consistent and detailed bug report format in the Excel tracking sheet ensured that all stakeholders had clear visibility into the QA status of each environment.

Reflections on Personal Growth and Learning Experiences:

Technical Proficiency: Expanded skills in QA for AI systems, covering identification of environment defects, regression testing, and structured bug documentation.

Problem-Solving Skills: Strengthened ability to analyze complex behaviors in RL environments, understand root causes, and communicate findings effectively to development teams.

Communication and Collaboration: Enhanced communication and teamwork abilities through daily interactions with developers, fostering efficient collaboration and timely resolution of issues.

2.5 Acquiring and Applying Knowledge

The theoretical knowledge acquired during academic studies, particularly in software testing methodologies, software engineering principles, and introductory AI and machine learning concepts, proved highly relevant in navigating the challenges of this internship.

Software Testing Life Cycle (STLC):

Academic knowledge of the Software Testing Life Cycle provided a structured framework for approaching QA tasks. The phases of requirement analysis, test planning, test case development, test execution, and defect reporting were all applied systematically during the internship.

Reinforcement Learning Fundamentals:

Understanding core RL concepts such as agents, environments, states, actions, rewards, and policies was critical to formulating meaningful test scenarios. The intern developed this knowledge through self-study and mentorship during the onboarding phase, directly applying it to identify behavioral anomalies in RL environments.

Development of Practical Skills:

Manual Testing Execution: Gained hands-on experience in designing and executing manual test cases for AI-driven environments, including exploratory, functional, and regression testing.

Bug Documentation: Developed expertise in writing clear, reproducible, and actionable bug reports, including identifying and documenting edge cases and unexpected system behaviors.

Technical Communication: Enhanced the ability to explain complex AI environment behaviors to developers in a precise and understandable manner, facilitating efficient resolution of identified defects.

Reflections on Hands-on Learning Experiences:

Practical Application: Applying theoretical concepts in real-world scenarios provided valuable insights into the practical relevance and applicability of academic knowledge. Hands-on experience in executing test scenarios, analyzing environment behavior, and maintaining bug records deepened understanding of QA concepts and their impact on software quality.

Immediate Feedback: Immediate feedback received during hands-on tasks allowed for quick validation of learning outcomes and identification of areas for improvement. The iterative learning process, combined with practical application, facilitated deep learning and retention of knowledge.

Professional Growth: Hands-on learning experiences substantially contributed to professional growth, equipping the intern with practical skills and confidence to tackle complex QA challenges in future endeavors.

2.6 Use of Modern Tools and Software

Throughout the internship, several tools and software were utilized to support QA activities:

Microsoft Excel:

The primary tool used for bug tracking and documentation. A structured Excel sheet was maintained to record all identified defects, including their descriptions, reproduction steps, severity levels, statuses, and resolution details. This provided a centralized, accessible record of all QA activities.

RL Environment Interface:

The RL environments developed by Scaler AI Labs (InterviewBit Software Services Private Limited) were directly interacted with during testing. Understanding and navigating the environment interface was essential to identifying behavioral deviations and reproducing reported defects.

Internal Communication Platforms:

Internal communication tools were utilized for daily coordination with the development team, sharing bug reports, clarifying environment specifications, and receiving updates on bug fix deployments for regression testing.

2.7 Use of Management Practices

During the internship, a robust framework of management practices was consistently applied to ensure the efficient execution of QA responsibilities and optimal project outcomes.

Communication:

Maintaining clear and proactive communication with the development team was central to the internship experience. Regular updates on testing progress, prompt reporting of newly identified bugs, and timely verification of resolved issues ensured a smooth and productive QA workflow. Clear and concise written communication in bug reports cultivated a culture of transparency and cooperation.

Teamwork:

Close collaboration with developers was fundamental to the success of QA activities. By working together to understand environment specifications, clarify behavioral expectations, and discuss potential fixes, the intern and development team operated as an integrated unit focused on delivering high-quality RL environments. Collaborative problem-solving fostered unity and boosted overall productivity.

Task Prioritization:

With multiple RL environments under concurrent development, effective task prioritization was essential. Testing efforts were organized based on the development stage, criticality, and complexity of each environment, ensuring that the most impactful areas received appropriate QA attention.

Reflections on Project Management Practices:

Clear and efficient communication emerged as fundamental to successful QA execution. The internship experience highlighted the importance of establishing open communication channels within the team to prevent misunderstandings, align expectations, and foster collaboration.

Collaboration proved to be a cornerstone of project success, enhancing problem-solving abilities and overall results. Maintaining a well-organized bug tracking sheet, adhering to consistent reporting standards, and sustaining open communication channels were key factors in the successful execution of QA responsibilities.

This experience reinforced the importance of structured and disciplined work practices in delivering quality outcomes and instilled a deep appreciation for the role of management practices in achieving professional excellence.

2.8 Design/Development of Solutions

The design and development of QA solutions during the internship followed a structured methodology tailored to the unique challenges of testing RL environments.

Requirement Analysis:

For each RL environment assigned for testing, the first step involved a thorough review of the environment's specification documents to understand the intended behavior, state space, action space, reward structure, and terminal conditions. This analysis formed the basis for all subsequent test design activities.

Test Scenario Design:

Based on the requirement analysis, comprehensive test scenarios were designed to cover a wide range of environment behaviors, including normal operation, boundary conditions, and edge cases. These scenarios were structured to systematically validate each aspect of the RL environment's functionality.

Test Execution and Bug Reporting:

Test scenarios were executed manually against the RL environment, with all observed deviations from expected behavior documented as bugs in the Excel tracking sheet. Each bug report was crafted with sufficient detail to enable developers to reproduce and resolve the issue efficiently.

Verification and Closure:

Following developer-implemented fixes, the intern conducted regression testing to verify that reported bugs had been resolved correctly. Verified bugs were marked as closed in the tracking sheet, and any regressions or incomplete fixes were re-reported for further developer action.

Analysis of Solution Effectiveness:

The structured QA approach adopted during the internship proved highly effective in ensuring the quality and reliability of RL environments. The systematic test design process ensured comprehensive coverage, while the detailed bug reporting facilitated efficient developer resolution. The iterative cycle of testing, reporting, fixing, and verifying resulted in measurably improved environment quality over time.

Enhancement of Software Efficiency:

Deployment of tailored QA solutions led to significant improvements in the quality and reliability of RL environments. By identifying and facilitating the resolution of defects early in the development cycle, the intern contributed to reducing the cost and effort associated with late-stage bug fixing.

2.9 Completion Status

Summary of Work Assignment Completion Status:

As the internship progresses towards its conclusion in May 2026, the following summarizes the status of ongoing and completed work assignments:

RL Environment Manual Testing:

Completed Tasks: Successfully completed comprehensive manual testing of all 12 assigned RL environments — Adobe Sign, Cal.com, Rover, MS Teams, Workable, Trello, Shopify, Buffer, Box, Google Calendar, QuickBooks, and SAP — progressing at a steady pace of one to two environments per week. Each environment underwent functional, exploratory, regression, and UI testing to ensure thorough quality coverage, with a final QA sign-off provided upon successful completion.

Outstanding Tasks: New RL environment modules continue to be introduced regularly, requiring ongoing testing engagement through the end of the internship period in May 2026.

Bug Documentation:

Completed Tasks: All identified defects have been systematically documented in the Excel bug tracking sheet with comprehensive details including reproduction steps, severity, and expected vs actual behavior.

Outstanding Tasks: Ongoing testing activities continue to generate new bug reports that are added to the tracking sheet in real time.

Bug Verification:

Completed Tasks: Successfully verified and closed all developer-resolved bugs reported to date, with regression testing confirming the correctness of implemented fixes.

Outstanding Tasks: New bug fixes continue to require regression verification as development progresses.

3. WEEKLY OVERVIEW

Onboarding and Learning Phase (Weeks 1–3):

Learning Objectives: The initial weeks of the internship were carefully structured to deliver comprehensive onboarding on the organization's products, RL environment architecture, and QA processes. The primary aim was to build a solid foundational understanding of RL systems and to cultivate familiarity with the tools and workflows used by the QA team.

Key Learnings:

RL Concepts: Gained an understanding of how Reinforcement Learning environments function, including the role of agents, states, actions, rewards, and terminal conditions. This foundational knowledge was essential for designing meaningful test scenarios.

Bug Tracking Process: Learned the organization's standardized bug tracking process, including how to document defects in the Excel sheet, the required fields and format, and the workflow for tracking bugs from identification through to resolution.

Exploratory Testing: Conducted initial exploratory testing on sample RL environments to develop intuition for typical environment behaviors and to identify areas requiring particular QA attention.

Active Testing Phase (Weeks 4 onwards):

Learning Objectives: Transitioning to active QA engagement, the focus shifted to applying the foundational knowledge developed during onboarding to real-world RL environment testing. The objective was to systematically test one to two RL environments per week, identify defects accurately, document them comprehensively, and verify their resolution in subsequent cycles.

Key Learnings:

Environment Pacing and Throughput: Developed a disciplined testing cadence of one to two environments per week. Simpler environments such as Adobe Sign, Cal.com, Rover, Buffer, and Trello were typically completed within a week each, given their relatively focused feature sets. More complex platforms such as Shopify, SAP, QuickBooks, MS Teams, and Google Calendar required two weeks of dedicated testing due to their extensive business logic, multi-module integrations, and large volume of edge cases. By the time of this report, all 12 assigned environments — Adobe Sign, Cal.com, Rover, MS Teams, Workable, Trello, Shopify, Buffer, Box, Google Calendar, QuickBooks, and SAP — have been successfully QA-tested and closed.

Test Case Design: Developed proficiency in designing comprehensive test cases that cover a wide spectrum of RL environment behaviors, including standard functional scenarios, boundary conditions, hidden workflows, and domain-specific edge cases.

Defect Identification: Honed the ability to distinguish between expected and unexpected application behaviors, identifying subtle defects — including UI inconsistencies, workflow mismatches, data handling errors, and evaluation logic issues — that could impact agent performance and learning outcomes.

Developer Communication: Developed strong cross-functional communication skills through regular interactions with developers, learning to articulate complex technical findings in a clear, concise, and actionable manner with structured reproduction steps.

Regression Testing: Gained practical experience in regression testing, ensuring that developer-implemented fixes resolved the reported defects

without introducing new issues into the environment, and providing final QA sign-off upon successful verification.

Time Management and Prioritization: Mastered effective time management strategies to handle the weekly environment throughput — including working additional hours when required for complex environments such as SAP and Shopify — while maintaining the depth and thoroughness of QA activities.

4. CONCLUSIONS

The internship at Scaler AI Labs (InterviewBit Software Services Private Limited) has been an immensely transformative and rewarding experience, characterized by rigorous learning, hands-on QA engagement, and invaluable insights into the world of Reinforcement Learning and AI quality assurance. Reflecting on the diverse experiences and lessons learned over the course of this internship evokes a profound sense of growth, achievement, and preparedness for future professional endeavors.

The foundational phase of the internship, dedicated to understanding RL environment concepts, organizational workflows, and QA processes, proved instrumental in laying the groundwork for all subsequent accomplishments. Building a thorough understanding of how RL environments function, and developing proficiency in the organization's bug tracking and reporting standards, established a robust foundation for navigating real-world QA challenges.

The practical testing phase provided an exceptional opportunity to apply theoretical knowledge in a dynamic, real-world AI development context. Progressing at a pace of one to two RL environments per week, all 12 assigned environments — Adobe Sign, Cal.com, Rover, MS Teams, Workable, Trello, Shopify, Buffer, Box, Google Calendar, QuickBooks, and SAP — were successfully QA-tested over the course of the internship. Systematically identifying defects in each environment, documenting them with precision, and collaborating with developers to ensure their resolution deepened both technical proficiency and professional maturity. The iterative cycle of testing, reporting, verifying, and providing final sign-off offered a comprehensive view of the software quality assurance process as applied to cutting-edge AI systems.

One of the most significant takeaways from this internship was the experience of bridging the gap between QA and development teams. Learning to understand complex RL environment behaviors deeply enough to explain them clearly to developers, and then verifying that fixes met the expected standards, underscored the critical role of effective communication and technical empathy in collaborative software development.

As the internship journey draws to a close, the experiences and lessons learned serve as guiding principles for future professional pursuits. The comprehensive exposure to AI quality assurance, practical testing methodologies, and cross-functional collaboration has not only enhanced technical skills but also nurtured essential soft skills such as communication, adaptability, and analytical thinking. These competencies collectively position the intern for meaningful contributions to the field of AI and software quality assurance in the years ahead.

REFERENCES

- [1] Sutton, R. S., & Barto, A. G. (2018). Reinforcement Learning: An Introduction (2nd ed.). MIT Press.
- [2] Desikan, S., & Ramesh, G. (2006). Software Testing: Principles and Practices. Pearson Education.
- [3] Kaner, C. (2000). How to Write a Good Bug Report. Florida Institute of Technology.
- [4] IEEE Software Engineering Standards. (2023). Quality Assurance in AI Systems.
- [5] Scaler AI Labs (InterviewBit Software Services Private Limited). (2025). Official Website. Available at: <https://www.scaler.com>